designed for up to 20 years of battery life. They communicate with a continuously active part of the mesh. Because Wi-SUN can deploy more routers than cellular systems, transmission distances can be shorter. This allows devices to transmit at lower power, further reducing energy consumption.

On co-existence with cellular networks

EFY. Can Wi-SUN coexist with private LTE or private 5G networks in the same utility deployments?

PB. Yes. Wi-SUN and private LTE or private 5G are often complementary rather than competing. Wi-SUN FAN operates in a different frequency band from LTE, which uses licensed spectrum. In many utility deployments, particularly in the United States, Wi-SUN FAN forms the wireless mesh connecting field devices, while the border router sends aggregated data to the utility office over LTE. Depending on the deployment, the border router may instead use fibre or Wi-Fi for this higher-bandwidth link.

This combination uses spectrum efficiently. Cellular is well suited to sending large data blocks over long distances but is less efficient for small packets such as smart meter readings. Wi-SUN FAN collects data from smart meters, switch reclosers, and other distribution equipment, aggregates it from hundreds of meters, and forwards it over cellular to the utility office.

On security and intelligence: post-quantum and AI

EFY. With quantum-safe cryptography becoming increasingly important, how is Wi-SUN preparing for the post-quantum era?

PB. Wi-SUN continues to evolve its specifications to address emerging security requirements. The next generation of the specification is already being developed, and utilities have identified post-quantum security as an important requirement. The Advanced Encryption Standard (AES) encryption currently used for securing Wi-SUN communications is already considered post-quantum safe by the US National Institute of Standards and Technology (NIST) for message encryption. The alliance is now focusing on strengthening the authentication process by incorporating post-quantum cryptographic algorithms. The current direction is to use the module-lattice-based key-encapsulation mechanism (ML-KEM) as a post-quantum key exchange mechanism, providing stronger protection against future quantum computing threats.

EFY. How is AI being used in Wi-SUN networks today?

PB. Artificial intelligence (AI) is being used across Wi-SUN networks for both edge intelligence and cybersecurity. Many smart meters now incorporate AI chips that perform edge processing, including energy usage disaggregation to identify the power consumption of individual appliances. For example, if a refrigerator's compressor begins to fail, the smart meter can detect the change in its energy signature and alert the consumer before a breakdown occurs. Utilities also use AI to monitor power quality and detect faults in overhead power lines. Wi-SUN's wireless mesh architecture supports these applications by enabling processing close to where the data is generated.

AI also strengthens cybersecurity. Because Wi-SUN is based on IPv6, every connected device has a unique IP address, allowing AI-based intrusion detection systems to monitor each device individually rather than only the gateway. If a cyberattack occurs, utilities can identify the exact device that has been compromised, providing greater visibility than technologies where monitoring is limited to the gateway.

On the road ahead: Openness, gaps and opportunities for India

EFY. What opportunities will the Wi-SUN ecosystem create for Indian embedded engineers and product developers?

PB. One of Wi-SUN's biggest strengths is that its specifications are open. Unlike some competing alliances or organisations that keep their specifications protected, Wi-SUN has contributed its specifications to the Institute of Electrical and Electronics Engineers (IEEE) and the International Electrotechnical Commission (IEC), allowing any manufacturer to obtain a copy and develop products that comply with international standards. Wi-SUN's role is to provide a testing and certification program, and we are committed to offering that at a reasonable cost so companies of all sizes, including those based in India as well as other parts of the world, can certify their products and participate in the ecosystem.

EFY. What gaps still exist in the Wi-SUN ecosystem, and how is the Alliance addressing them?

PB. There is still some scepticism among certain smart meter vendors about implementing Wi-SUN, but we believe training and education will help address those concerns. We are also continuously expanding and refining the Wi-SUN specifications while working closely with complementary industry organisations through liaison agreements. For example, the Device Language Message Specification (DLMS) User Association's DLMS/COSEM (companion specification for energy metering) protocol has been specified by the Bureau of Indian Standards as a smart meter application protocol, and the Wi-SUN Alliance has a liaison with that organisation. Similarly, Lightweight M2M has been specified, and Wi-SUN works with the Open Mobile Alliance (OMA) to ensure compatibility. These collaborations help us deliver a complete protocol specification that integrates complementary technologies. EFY